1 Methods

1.1 Pre-processing and few-shot classification

[Placeholder for the existing Sect. 2.1 text – rendering of the spectra, the plotting conventions, and the few-shot classification setup. Delete this subsection when transplanting.]

1.2 Agentic construction and two-stage neuro-symbolic adjudication

The classifier described in Sect. 1.1 was not hand-tuned. Both the system instruction, the contrastive few-shot reference set, and the two-stage inference and adjudication pipeline were produced by a multi-agent system built on the Google Agent Development Kit (ADK), in which a deterministic orchestrator chains a set of specialised sub-agents, each responsible for one reproducible step of the experiment (Fig. 1). The implementation is publicly available at `github.com/turanbulmus/cda-dust-analyzer`.

Literature grounding. The system instruction was not written from the instrument description alone. A research agent ingested the full text of the key CDA and Enceladus publications and was tasked with extracting the physical and chemical constraints that bear on classification — the mass resolution of $m/\Delta m \approx 10$–50, the $t = t_0 + a\sqrt{m/z}$ time-of-flight mapping ($a \approx 0.473\,\mu$s) and the resulting trigger shift of up to ± 50 channels, the hydronium cluster combs of icy grains, the Mg-rich and Fe-depleted silicate cations, the 12–13 u carbon spacing of macromolecular organics, and the hydroxide $Na^+(NaOH)_n$ ($63, 103, 143, 183$ u), carbonate $Na^+(Na_2CO_3)$ (129 u), and orthophosphate $Na^+(NaPO_3)$ / $Na^+(Na_2HPO_4)$ / $Na^+(Na_3PO_4)$ ($125, 165, 187$ u) cluster series of salt-rich Type 3 grains — and with translating them into explicit, falsifiable decision rules. A second, adversarial critic agent audited that proposal for overfitting, physical degeneracies (such as the $\Delta t = 1.45\,\mu$s spacing between H_3O^+ at 19 u and $H_3O^+(H_2O)_2$ at 55 u mimicking the $1.49\,\mu$s $Na^+ \rightarrow Na^+(NaOH)$ spacing), and recall-degrading over-rejection hazards (such as penalising hydrated $Na^+(H_2O)$ at 41 u as potassium K^+ at 39 u). A third referee agent adjudicated the exchange and synthesised only the surviving, physically verified rules into the final system instruction. Every criterion in the prompt therefore encodes published CDA physics with traceable provenance.

Contrastive few-shot construction. Each labelled training spectrum was rendered with the identical plotting routine used at inference and described in natural language by the model. Crucially, the model wrote each description while simultaneously viewing one reference example of every other class, so that the resulting one- or two-sentence annotation states what *distinguishes* that spectrum from its neighbours rather than what it contains in isolation. These per-spectrum descriptions were then consolidated class by class in an autonomous hierarchical loop: at each level the agent merged the descriptions it had been given, reported how many genuinely distinct morphologies remained, and decided for itself whether a further merge was warranted or whether the surviving patterns were fundamental to the class. The loop ran for at most five levels and converged to between one (Class 3-P) and four (Class 1) canonical patterns per class. The final reference set comprised 16 examples per class (four for the sparsely populated Class 3-P) in the multiclass survey, and 18 stratified exemplars (nine positive and nine look-alike negatives spanning `3-Cl`, `3-Car`, `3-P`, `3-K`, `3-KNa`, and Types 1–2) in the dedicated Type 3 subtype workflows.

Cost-bounded Stage-1 inference and implicit caching. Because roughly two thirds of the general survey is instrumental noise, submitting every sweep to the model would have been wasteful. A deterministic Stage-1 gating pre-filter therefore screens each spectrum before rendering: a sweep is bypassed as noise only if its 15-channel rolling mean spans less than 0.075 in normalised amplitude, or if it contains no peak of height 0.20 and prominence 0.08 outside the digitizer artefact regions (channels < 20 and > 620) while carrying a target impact charge below 1 fC. Calibrated against the training set so that the weakest genuine chemical grain (smoothed range 0.080) is preserved with zero false negatives, this filter removes 23% to 65% of inference calls depending on archive subset. Surviving spectra were dispatched as asynchronous Vertex AI batch jobs prepared via Apache Beam; because every request shares an identical few-shot exemplar prefix (~ 26 k tokens) followed by a single target spectrum (~ 2.9 k tokens), 86%–91% of input tokens hit automatic implicit context caching, reducing full-archive Stage-1 inference across 18,305 labelled spectra to \$112.49.

Stage-2 mass-calibrated neuro-symbolic and VLM adjudication. While Stage-1 time-of-flight screening achieves high sensitivity (90.2% recall on `3-OH` across 18,305 spectra; 100% recall on held-out `3-P`), raw time-domain traces suffer from per-spectrum trigger jitter (t_0) and stretch factor (a) variations that blur the 4 u separation between phosphate $Na^+(NaPO_3)$ (125 u) and carbonate $Na^+(Na_2CO_3)$ (129 u), as well as between carbonate 129 u and hydroxide $Na^+(NaOH)_3$ (143 u). All Stage-1 positive candidates (64–382 spectra per survey, adding $<$ \$1.50 per run) are therefore routed to a Stage-2 specialist adjudicator. First, a physics-constrained two-anchor calibration solves $(\hat{t}_0, \hat{a})$ explicitly for each candidate from the 90%-contour centroid of Na^+ (23.0 u) and the sharp $Na^+(NaOH)$ (63.0 u) apex, mapping the horizontal axis directly to mass $m/z = [(t - \hat{t}_0)/\hat{a}]^2$. Second, the spectrum is re-rendered as a dual-panel mass-calibrated figure pairing the full 10–200 u view with a zoomed

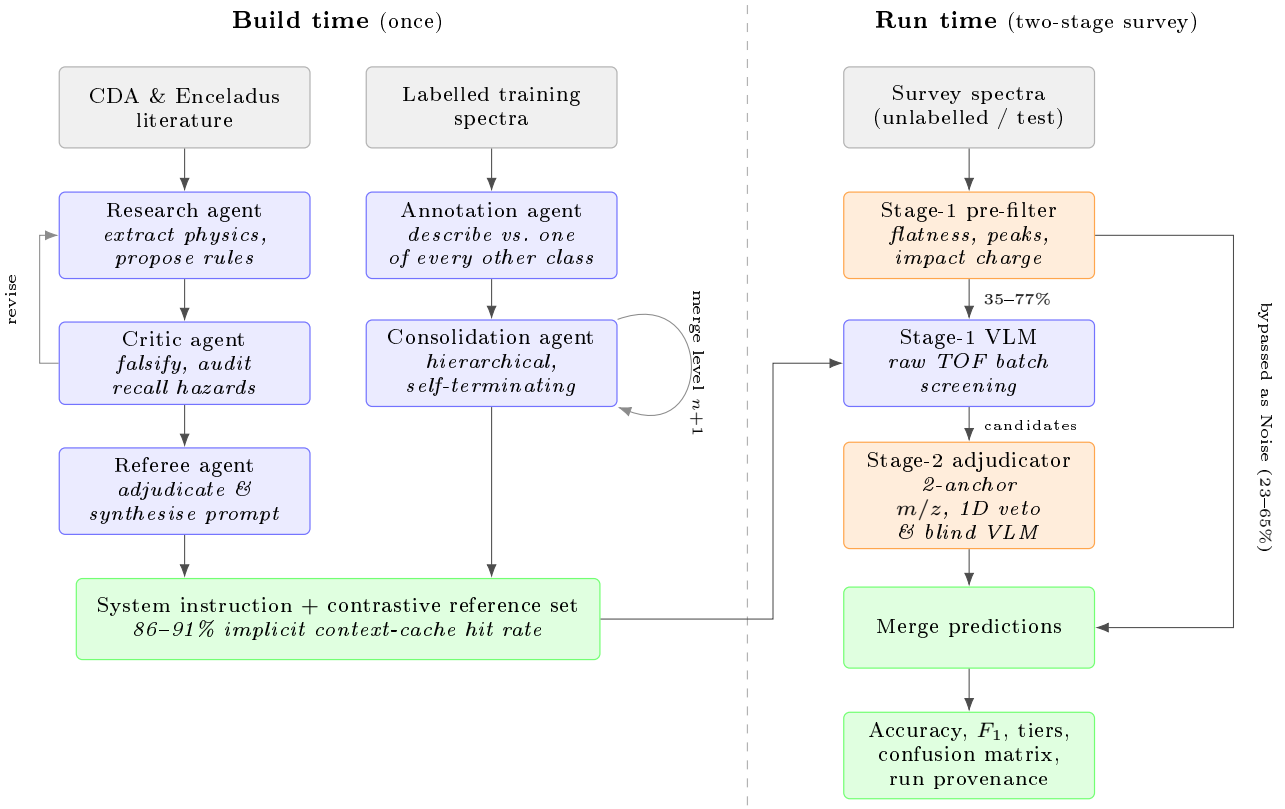


Figure 1: Two-stage agentic workflow used to build, run, and adjudicate the classifier. At build time (left), a three-agent loop (Research → Critic → Referee) derives and falsifies physical decision rules from the published CDA literature, while in parallel reference spectra are annotated contrastively and consolidated into canonical class patterns. At run time (right), a deterministic Stage-1 pre-filter bypasses obvious noise (23–65%) before Stage-1 Gemini VLM batch screening on raw time-of-flight plots. Positive salt-subtype candidates (`3-OH`, `3-P`) are then routed to the Stage-2 specialist adjudicator, which performs two-anchor mass calibration $(\hat{t}_0, \hat{a})$ on Na^+ (23 u) and Na^+(NaOH) (63 u), renders dual-panel mass-calibrated plots (10–200 u and 70–195 u), and applies 1D mass-window physics vetoes alongside a leak-free blind specialist VLM before merging predictions and logging provenance.

70–195 u anion diagnostic panel overlaid with colour-coded vertical mass gridlines. Third, a two-tier neuro-symbolic gate combines 1D baseline-subtracted mass-window prominence telemetry with a blind Stage-2 specialist VLM (`gemini-4-argon` inspecting the dual-panel mass plot and raw peak heights without threshold leakage). On the 18,305-spectrum labelled archive, the 129 u carbonate and water-comb physics vetoes cut `3-OH` false positives by 77.8%–88.9% (27 → 6 and 3), raising precision from 57.8% to 85.4%–91.7% (F_1 = 0.854–0.857). On the 377 Stage-1 `3-P` batch false positives from 17,146 spectra, a parameter-free physical sign screen ($p_{125} \geq p_{129}$) retains 100% (9/9) of held-out catalogued and mixed-phase `3-P` grains while rejecting 83.8%–93.1% of false positives, and the blind Stage-2 specialist VLM rejects 100% (377/377) of Stage-1 false positives while retaining 100% (8/8) of canonical catalogued `3-P` grains (100% batch precision, $F_1 = 1.00$).

2 Results

[Placeholder for the benchmarking results. Delete this section when transplanting.]